

520 Full Rebuild Checklist — Run 17_1

Purpose: Printable step-by-step guide to run a **full end-to-end rebuild** of the Army Cox pipeline using `pipeline_config_17_1` (Run 1: z-scored TB ratio x pool minus mean).

Notebook: `talent/talent_pipeline/520_pipeline_cox_working.ipynb`

Also valid path: `talent_pipeline/520_pipeline_cox_working.ipynb` (repo-root symlink — same files)

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Before you start

- ☐ Jupyter working directory is `talent/talent_pipeline/` (checkpoints save to `./running_vars/` next to the notebook).
- ☐ You have enough disk/RAM for a full run (Cells 4–6 are the heaviest; see `520_PIPELINE_COX_OVERVIEW.md` § Memory).
- ☐ Ground truth code is `talent/talent_pipeline/`, not a stale copy under `Army_AWS_download/` unless you just merged from AWS.

The three control knobs (where to edit what)

What you want	File	Where
Pick run recipe (17_1)	<code>520_pipeline_cox_working.ipynb</code>	Cell 0 → <code>pip_config_file = 'pipeline_config_17_1'</code>
Full vs partial rebuild	<code>pipeline_config.py</code>	§1 → <code>CELL1</code> ... <code>CELL12_8</code> flags (<code>True</code> = run)
Run 1 filters / model / plot names	<code>pipeline_config_17_1.py</code>	Usually leave as-is for standard Run 1

Not automatic: Creating `pipeline_config_XX_Y.py` does **not** run it. Only `pip_config_file` in Cell 0 selects the override.

Output naming: Set by `plot_prefix` in `17_1`'s `RUN_PROFILE` (default `'run1'` → e.g. `run1_cr_z_tb_ratio_fwd_snr`).

Phase A — Upstream data (if feathers missing or stale)

Run these **before** 520 if `./running_vars/` lacks the inputs.

A1. Snapshot base — `502_working.ipynb`

- ☐ Open `502_working.ipynb` (same folder as 520).
- ☐ Run through to produce `df_502_base.feather` in `./running_vars/`.

A2. OER enrichment — `512_oer_int_working.ipynb`

- ☐ Open `512_oer_int_working.ipynb`.
- ☐ Run Cells 0–3 (minimum) to produce `df_oer_enriched.feather` in `./running_vars/`.
- ☐ 520 Cell 4 **requires** this file; it will error with instructions if missing.

A3. Verify inputs exist

- ☐ `./running_vars/df_502_base.feather`
- ☐ `./running_vars/df_oer_enriched.feather`

Phase B — Base config: turn on full rebuild

File: `talent/talent_pipeline/pipeline_config.py`

Section: `=== 1. CELL EXECUTION FLAGS ===` (top of file)

Set every flag below to `True` for a full rebuild:

Phase 1 — Load & prep

- ☐ `CELL1 = True` — Load raw snapshots
- ☐ `CELL2 = True` — Basic filtering
- ☐ `CELL3 = True` — Year group (yg)

Phase 2 — OER & pools

- ☐ `CELL4 = True` — OER merge + individual fwd/bwd metrics
- ☐ `CELL5_POOL_MEANS = True` — Pool means / minus-means / sizes
- ☐ `CELL6_POOL_RANKS = True` — Pool ranks / percentiles / z-scores

Phase 3 — Variables

- ☐ `CELL7 = True` — Time-varying variables (incl. `div_name` when 17_1 enables it)
- ☐ `CELL8 = True` — Static variables

Phase 4 — Filter

☐ CELL9 = True — Advanced filtering (17_1: division filter on)

Phase 5 — Cox

- ☐ CELL10 = True — Cox interval prep
- ☐ CELL10_5 = True — Standardize + interactions + quadratics
- ☐ CELL10_5S = False — Splines (leave False unless experimenting)
- ☐ CELL11 = True — CR/CIF plots
- ☐ CELL12 = True — Cox models

Cell 12 sub-steps

- ☐ CELL12_1 = True — Prepare Cox data
- ☐ CELL12_2 = True — Static model
- ☐ CELL12_3 = True — Full model
- ☐ CELL12_4 = True — Model comparison / signal ratios
- ☐ CELL12_5 = True — Competing risks
- ☐ CELL12_6 = True — Partial effects
- ☐ CELL12_7 = True — Interaction 3D surfaces
- ☐ CELL12_8 = True — Model-based curves

Optional pre-run

☐ RUN_503 = False — Set True only if you need to rebuild prestige UIC lists (py_503_hierarchies.py) before Cell 7.

Note: CELL0 in config is legacy; the notebook Cell 0 block for lookup tables is commented out — ignore for normal runs.

Save pipeline_config.py .

Phase C — Select Run 17_1 in the notebook

File: 520_pipeline_cox_working.ipynb

Cell: Cell 0 (imports + config load)

☐ Set:

```
pip_config_file = 'pipeline_config_17_1'
```

(module name only — no .py)

☐ Run Cell 0. Confirm printed module name is pipeline_config_17_1 .

Cell 1 inline option (same notebook, first data cell)

☐ Confirm load_from_502 = True (recommended — uses df_502_base.feather).

Phase D — Execute 520

- ☐ Restart kernel (optional but clean for a full rebuild).
- ☐ Run Cell 0, then run all cells through Cell 12 (or Run All).
- ☐ Watch Cell 4 for OER load success; Cells 5–6 for pool metrics; Cell 9 for filter counts.

Expected checkpoint files (./running_vars/)

- ☐ df_pipeline_01_raw.feather
- ☐ df_pipeline_02_base.feather
- ☐ df_pipeline_03_base.feather
- ☐ df_pipeline_04a_basic_metrics.feather
- ☐ df_pipeline_05_pool_means.feather
- ☐ df_pipeline_06_pool_ranks.feather
- ☐ df_pipeline_07_time_varying.feather
- ☐ df_pipeline_08_combined.feather
- ☐ df_pipeline_09_filtered.feather
- ☐ df_pipeline_10_cox_ready.feather
- ☐ df_pipeline_10_5_cox_zscored.feather
- ☐ df_pipeline_11_cox_analysis.feather
- ☐ df_pipeline_12_01_prepared.feather

Phase E — Verify outputs

Plots

☐ `./cox/cox_plots/` — CR/CIF figures prefixed with `run1_` (e.g. `run1_cr_z_tb_ratio_fwd_snr`, pool minus mean, interaction).

Model / diagnostics

☐ `./cox/cox_results/` (or paths under `./cox/` per your config) — coefficients CSV, interaction 3D PNG (`interaction_3d_star_pool_interaction_...`).

Sanity checks

- ☐ Cell 11 completed without “missing variable” errors for `z_tb_ratio_fwd_snr`, `z_pool_minus_mean_snr_fwd`, `star_pool_interaction`.
- ☐ Cell 12.3 full model fitted; 12.7 saved at least one 3D surface.

Phase F — After successful rebuild (optional)

For **faster iteration** next time (Cox/plots only):

- ☐ In `pipeline_config.py`, set `CELL1` through `CELL6` = `False` (reuse feathers through Cell 6).
- ☐ Keep `CELL7` – `CELL12` as needed (re-run from 7 if you changed division/filter config in 17_1).

What Run 17_1 does (no edits needed for default Run 1)

File: `pipeline_config_17_1.py`

Setting	Value
<code>RUN_SCALE</code>	<code>'z'</code> (z-scored model columns)
Model TV vars	TB ratio SNR + pool minus mean SNR
Interaction	<code>star_pool_interaction</code>
<code>plot_prefix</code>	<code>'run1'</code>
Quadratics	Both TB ratio and pool minus mean
Division	<code>DIVISION_CONFIG</code> on; <code>div_name</code> in model; division filter in Cell 9
CR plots	Explicit list (<code>USE_OVERRIDE_CR_PLOTS = True</code>)

Edit `17_1` only when you want different filters, variables, or `plot_prefix` — not for “run again from scratch.”

Troubleshooting (quick)

Symptom	Check
<code>df_oer_enriched.feather not found</code>	Run 512 Cells 0–3
<code>df_502_base.feather not found</code>	Run 502, or set <code>load_from_502 = False</code> and use DB path in Cell 1
Wrong run / wrong plots	Cell 0 <code>pip_config_file</code> ; re-run Cell 0 after change
Config edit ignored	Re-run cell that calls <code>reload_pipeline_config()</code> , or restart kernel + Cell 0
Cells skipped unexpectedly	<code>pipeline_config.py</code> CELL flags — must be <code>True</code> for full rebuild
Empty interaction plot range	17_1 <code>interaction_percentile_range</code> ; winsorize settings in <code>RUN_PROFILE</code>

Related docs (deeper reference)

Doc	Topic
<code>520_PIPELINE_COX_OVERVIEW.md</code>	Full pipeline architecture
<code>PIPELINE_CONFIG_OVERVIEW.md</code>	CELL flags + RUN_PROFILE
<code>PIPELINE_RUN_ORDER.md</code>	Checkpoint chain one-liner
<code>README_Talent_Layout_Symlinks_And_AWS_Export.md</code>	AWS vs local tree
<code>CR_AND_HR_FOR_DUMMIES.md</code>	Reading CIF vs HR plots
<code>Presentation_Interpretation_Run1_Slides_5-20.md</code>	Run 1 figure interpretation

One-page summary

1. 502 → df_502_base.feather (if needed)
2. 512 → df_oer_enriched.feather (if needed)
3. pipeline_config.py → CELL1...CELL12_8 = True
4. 520 Cell 0 → pip_config_file = 'pipeline_config_17_1'
5. 520 Cell 1 → load_from_502 = True
6. Run All 520
7. Check ./cox/cox_plots/run1_* and ./running_vars/df_pipeline_* checkpoints
8. Later: CELL1-6 = False for Cox-only reruns

Print this checklist and check boxes as you go.